

Project 1 Report: Advanced EDA & Feature Engineering

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1. Introduction

Exploratory Data Analysis (EDA) is a crucial step in the data science workflow that involves understanding, cleaning, and preparing raw data before applying machine learning algorithms. Real-world datasets often contain missing values, outliers, inconsistent data formats, and categorical variables that must be transformed into numerical representations. Without proper preprocessing, machine learning models may produce inaccurate or biased predictions.

The objective of this project was to transform a raw retail sales dataset into a clean, structured, and machine-learning-ready dataset by performing data preprocessing, outlier treatment, feature engineering, and categorical variable encoding.

2. Project Objectives

The primary objectives of this project were to:

- Load and understand the dataset.
- Identify and handle missing values.
- Detect and treat outliers using statistical techniques.
- Engineer new predictive features from existing data.
- Convert categorical variables into numerical form.
- Save the cleaned dataset for future machine learning tasks.

3. Dataset Description

The dataset contains 1,200 retail transaction records with 14 original features representing customer orders.

Original Dataset Attributes

Feature	Description
OrderID	Unique order identifier
Date	Date of order
CustomerID	Unique customer identifier
Product	Product purchased
Quantity	Number of items purchased
UnitPrice	Price per item
ShippingAddress	Customer shipping address

PaymentMethod	Payment method used
OrderStatus	Current order status
TrackingNumber	Shipment tracking number
ItemsInCart	Number of items in shopping cart
CouponCode	Applied coupon code
ReferralSource	Marketing referral source
TotalPrice	Total order price

4. Methodology

The following preprocessing steps were performed to prepare the dataset.

Step 1: Load Data

The first step involved loading the retail dataset into Python using the Pandas library.

The dataset was imported from a CSV file into a DataFrame, allowing efficient manipulation and analysis of the data.

After loading the dataset, the following exploratory functions were used:

- `head()` to display the first five records.
- `tail()` to display the last five records.
- `shape` to determine the number of rows and columns.
- `columns` to list all feature names.
- `dtypes` to identify data types.
- `info()` to summarize dataset structure.

This initial exploration helped in understanding the dataset before preprocessing.

Step 2: Check Missing Values

Missing values are common in real-world datasets and can negatively affect machine learning models.

The dataset was examined using:

```
df.isnull().sum()
```

This function returned the number of missing values in each column.

The analysis showed that only the **CouponCode** column contained missing values, while all other columns were complete.

Identifying missing values is an essential preprocessing step because many machine learning algorithms cannot process incomplete data.

Step 3: Impute Missing Values

"The dataset was examined for missing values. No missing values were found in numerical features; therefore, statistical imputation methods (Mean, Median, and KNN) were not required. The only missing values occurred in the categorical CouponCode column and were imputed with the value NoCoupon to preserve their business meaning."

Step 4: Detect Outliers

Outliers are observations that significantly differ from the majority of the dataset and may distort statistical analysis and machine learning models.

The Interquartile Range (IQR) method was used to identify outliers.
The numerical columns analyzed were:

- Quantity
- UnitPrice
- ItemsInCart
- TotalPrice

The analysis revealed that only **TotalPrice** contained **eight outliers**, while the remaining numerical features contained no significant outliers.

Step 5: Neutralize Outliers

After detecting the outliers, they were treated using Winsorization (Capping).

Instead of deleting the outliers, extreme values were replaced with the calculated lower or upper IQR boundaries.

Winsorization was selected because:

- It preserves all observations.
- It reduces the influence of extreme values.
- It minimizes distortion in machine learning models.
- It is well suited for retail transaction datasets where unusually large purchases may still represent valid business events.

Box plots generated before and after Winsorization confirmed that the extreme values had been successfully capped while maintaining the overall data distribution.

Step 6: Create New Features

Feature engineering was performed to generate additional predictive variables from existing data.

The following features were created:

OrderMonth

Extracted the month from the Date column.

This feature captures seasonal purchasing patterns.

OrderYear

Extracted the year from the Date column.

This feature allows yearly sales trend analysis.

AveragePricePerItem

Calculated using:

$\text{AveragePricePerItem} = \text{TotalPrice} / \text{Quantity}$

This represents the average amount spent on each purchased item.

CartUtilization

Calculated using:

$\text{CartUtilization} = \text{Quantity} / \text{ItemsInCart}$

This feature measures how much of the shopping cart resulted in completed purchases.

DiscountUsed

A binary feature indicating whether a promotional coupon was applied.

Values were assigned as:

1 = Coupon used

0 = No coupon used

Feature engineering increases the amount of useful information available to machine learning algorithms and often improves predictive performance.

Step 7: Encode Categorical Variables

Machine learning algorithms require numerical input and cannot process textual categorical variables directly.

The following categorical columns were encoded:

- Product
- PaymentMethod
- OrderStatus
- CouponCode
- ReferralSource

One-Hot Encoding was applied using the Pandas `get_dummies()` function.

The parameter:

`drop_first=True`

was used to remove one category from each feature.

The removed category becomes the reference category, preventing redundant variables and avoiding the dummy variable trap, which can negatively affect certain machine learning models.

The resulting encoded columns contained binary values (0 or 1), making the dataset fully numerical and suitable for machine learning.

Step 8: Save the Cleaned Dataset

After completing all preprocessing and feature engineering tasks, the cleaned dataset was exported to a new CSV file.

This preserved the transformed dataset for future machine learning experiments without requiring the preprocessing steps to be repeated.

The cleaned dataset contains:

- No missing values.
- Treated outliers.
- Engineered predictive features.
- Encoded categorical variables.
- Numerical features ready for machine learning